

REPORT MILESTONE 2: PLN

STUDENTS:

1. Francisco Gil Valverde
 2. Gastón Fernández Rodríguez
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1. General description of the work

In this work, we study the detection of offensive language in the **HateBR** corpus, which contains Portuguese comments labeled as *offensive* or *not offensive*. The goal of **Milestone 2** is to develop and compare different **machine learning classifiers**, combining:

- Standard text representations (**TF-IDF** and **GloVe embeddings**), and
- A set of **linguistic features** designed from the ideas used in the rule-based system of **Milestone 1**.

At the end, we choose one **global winning model** using a validation set, and we evaluate it only once on a separate test set that was not used during development.

2. Train / validation / test split and selection metric

The HateBR corpus is split into three parts:

- **Train**: used to fit the model parameters and to run the internal cross-validation of GridSearchCV.
- **Validation**: used only to compare the six final models (two text representations $\times$ three algorithms) and to choose the **global winning model**.
- **Test**: used only once at the end to evaluate the winning model and to get the final performance.

Even though the dataset is **balanced between classes**, we still use stratify in all splits to keep the same proportion of offensive and non-offensive comments in train, validation and test.

In this context, we use **accuracy** as the main metric in GridSearchCV, because with balanced classes it is easy to interpret and reliable. We also report **macro F1**, which in our case is very similar to accuracy. This shows that the model behaves in a balanced way for both classes.

3. Feature scaling and relative importance of information

In the numeric preprocessing, we only scale the **linguistic features** using StandardScaler. We do **not** scale TF-IDF or GloVe embeddings:

- **TF-IDF** is already normalized by definition, and scaling it again could change the meaning of its values.
- **GloVe embeddings** are pre-trained vectors. Their space already encodes semantic relations, and changing their scale can damage this structure.
- **Linguistic features**, on the other hand, have **very different ranges** (counts, ratios, etc.), so we standardize them to avoid that features with large numeric values dominate the others.

After scaling, all linguistic features have similar numeric ranges, but this **does not mean** that the model gives them the same importance. In linear models such as Logistic Regression or linear SVM, each feature has a **coefficient** learned during training. The model can learn, for example, that “having at least one offensive word” is much more informative than small changes in comment length, and so it gives a larger weight to that feature.

In other words, the model automatically decides which features are more important for separating offensive from non-offensive comments.

4. Text representation: TF-IDF vs GloVe and cleaning decisions

We use two main ways to represent the text:

1. **TF-IDF (Model A)**

TF-IDF converts each comment into a sparse vector, where each dimension is a term (unigram or bigram) and the value shows how characteristic that term is for the document and for the corpus. In our case we use:

- `ngram_range=(1, 2)` to capture both single words and short expressions.
- `lowercase=False` to keep uppercase letters, which can show emphasis or aggressiveness.
- `max_features=10000` to control the dimensionality.

The cleaning for TF-IDF is **very light**: we only remove URLs and normalize spaces. We keep uppercase letters, punctuation and emojis. This helps to preserve surface signals that are important in offensive language (for example, many exclamation marks, words in ALL CAPS, etc.).

2. **GloVe embeddings (Model B)**

With GloVe, each word is represented by a pre-trained dense vector, and each comment is represented by the **average of the vectors of its words**. This representation captures semantic similarity between words, so similar insults or negative expressions are close in the vector space.

For GloVe we use a stronger cleaning: we remove URLs, convert text to **lowercase**, replace certain emojis with equivalent Portuguese words, remove non-alphabetic characters and normalize spaces. We do this because the GloVe vocabulary only contains “clean” word forms and is case-sensitive. If we do not clean the text, many tokens would not have an embedding.

In summary, **TF-IDF** uses minimal cleaning to keep surface information, while **GloVe** needs stronger cleaning to match the embedding vocabulary and to create good average vectors.

5. Linguistic features and connection to the rule-based system

Besides text representations, we add a set of **linguistic features** that complement TF-IDF and GloVe. We extract these features with spaCy from a version of the text where we only remove links and extra spaces, but we keep the original style.

The features include:

- **Length and style**: number of tokens, ratio of uppercase characters, number of exclamation marks and question marks.
- **Part-of-speech distribution**: proportion of adjectives, nouns and verbs (computed over the sum of these three categories).
- **Pragmatic clues**: number of emojis, hashtags, user mentions, and second person pronouns.
- **Syntactic patterns**: counts of patterns like NOUN + ADJ or 2nd-person PRONOUN + ADJ, which are common in direct insults.
- **Offensive lexicon**: number of tokens that belong to a list of offensive words identified in Milestone 1.

These features come directly from the **rule-based system** built in the first part of the project. In Milestone 1, the system used rules like “if the comment contains a word from the offensive list and many exclamation marks, then it is probably offensive”. In Milestone 2, we **reuse the same ideas** but now as numeric features. The machine learning model can then learn how to combine them and how important each of them is.

In this way, we integrate **linguistic knowledge** into a supervised model.

6. Project workflow and experimental structure

The complete workflow of the project is:

1. **Load the HateBR corpus** and keep the relevant columns (id, text, label).
2. **General cleaning** for linguistic features (remove links and extra spaces, keep everything else).
3. **Specific cleaning for each representation**:
 - A “soft” version of the text for TF-IDF (keeping uppercase, emojis and symbols).
 - A “clean” version for GloVe (lowercase, no symbols, emojis converted to words).

4. **Extract linguistic features** with spaCy and build a DataFrame with all numeric variables.
5. **Split the data** into train, validation and test, using stratify to keep balanced classes.
6. **Model A:** combine TF-IDF with linguistic features.
7. **Model B:** combine GloVe average embeddings with linguistic features.
8. **Select the global winning model** on the validation set and evaluate it only once on the test set.

With this design, we can compare in a fair way the effect of each text representation (TF-IDF vs GloVe), while keeping the same extra features and the same algorithms.

7. Machine Learning models used

For both Model A and Model B we use three classic supervised algorithms:

- **Logistic Regression**
- **Linear SVM (LinearSVC)**
- **Random Forest**

We choose these algorithms because:

- They are **standard baselines for text classification** and work well with high-dimensional data.
- Logistic Regression and linear SVM are very efficient with sparse matrices like TF-IDF and allow us to interpret the role of each feature through their coefficients.
- Random Forest is a non-linear model that can capture more complex interactions between features, giving a different view on the data.

We do not use more complex models (such as deep neural networks), because the main goal of Milestone 2 is to compare representations and features, and to keep the models relatively simple and explainable.

8. Results and winning model

For each combination (TF-IDF/GloVe $\times$ algorithm), we run a **GridSearchCV** on the training set to select the best hyperparameters. Then, we evaluate each best model on the **validation set**.

The results show that:

- Models based on **GloVe + linguistic features** achieve **better performance** than models based on TF-IDF + linguistic features.
- For both representations, **Logistic Regression** and **linear SVM** clearly outperform Random Forest, which agrees with common results in text classification.

The **best model on the validation set** is the **linear SVM with GloVe + linguistic features**, which we select as the **global winning model**. We then evaluate this single model on the **test set**. The results show similar performance for both classes (offensive and non-offensive), with accuracy and macro F1 around **0.82**. This means that the system classifies offensive and non-offensive comments with a good and balanced performance, and it meets the objectives defined for Milestone 2.